Adil Malik

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EDUCATION

Duke University **2018 — 2023**

Bachelor of Sciences in Computer Science and Electrical and Computer Engineering - **Double Major** **CGPA: 3.54**

Academic Honors: T.R Mullen Jr Scholarship for outstanding Engineer - $45,000 annually

PROFESSIONAL EXPERIENCE

Alarm.com, *Software Engineer II* | Virginia, USA **Sept 2025 — Dec 2025**

- Resolved device-versioning issue where v2 control panels were improperly provisioned as v1 by re-architecting C++ flow logic to treat v2 as distinct entity & restore over-the-air auto-updates to 5000+ devices installed in residential properties
- Identified & fixed compounding IMEI registration defect where React frontend input was incorrectly stored in backend by tracing fault across UI, C++ backend, & SQL database, which helped provide patch to 16,000+ active control panels

Capital One, *Software Engineer* | Virginia, USA **Aug 2023 — Sept 2025**

- Created automated migration engine, architecting AWS infrastructure from the ground up using Glue, Fargate, Kafka, DynamoDB, & Lambda to generate OneLake metadata & Iceberg/Unity tables, to establish a single source of truth
- Migrated 30 PB of data from Snowflake to Iceberg to eliminate duplicate storage & reduce annual data storage cost by $25M+
- Built observability backbone for migration engine in NewRelic & led cross-team standardization of instrumentation to trace request end-to-end across internal logic & APIs; reducing troubleshooting time for reported issues from days to minutes
- Automated SQL query-runtime benchmarking between OneLake & Snowflake to reduce turnaround from 3 days to 4 hours

Capital One, *Technology Internship Program* | Virginia, USA **May 2022 — Aug 2022**

- Led company-wide transition from Tableau to QuickSight in collaboration with Enterprise Product & Production teams by redesigning 30+ legacy dashboards to deliver actionable KPIs & insights to management on performance of ~1500 teams
- Optimized SQL queries over 200M+ records to reduce refresh time from 2 min. to 45 sec. & ensure integration at scale
- Built productivity-monitoring dashboards using advanced QuickSight features based on feedback by senior management

Google, Inc., *Software Engineering Intern* | California, USA **May 2021 — Aug 2021**

- Incorporated external PyTorch Mobile library into Google's codebase, enhancing AI compatibility for Android platforms
- Developed inference calculator to run PyTorch models within MediaPipe to establish CPU-based execution for Tensor chip
- Integrated MediaPipe with the Tensor chip's accelerated edge TPU computing via NNAPI, boosting inference speed by 20x & FPS by 500% & enabling end-to-end PyTorch model support on-device

LEADERSHIP EXPERIENCE

Tech Development Program Council, *Site Lead,* Capital One **Aug 2023 — Jan 2025**

- Directed 4 committees of 30+ members & managed $200K+ budget to deliver 40+ events for new-grad employees
- Spearheaded organization of 2024 Capital One TDP Hackathon attended by 300+ participants, across 5 office locations
- Launched Clubs initiative by reallocating existing budget & building a governance & access-control framework from scratch that enabled 1,200+ TDP employees to lead, fund, & manage their own interest groups
- Leveraged measured gains in employee satisfaction to secure $20,000 annual Clubs budget approval for a 5-year period

Duke Jumma4All, *Co-Founder*, Duke University **May 2022 — May 2023**

- Awarded LGBTQ+ Leaders to Promote Community Award by Duke's Center for Multicultural Affairs for establishing group that fostered a queer-affirming, anti-racist & gender-inclusive space focused on intersection of LGBTQ & Muslim identity

Duke Muslims Students Association, *Civic Engagement and Outreach Chair*, Duke University **Feb 2020 — Feb 2021**

- Partnered with a national charity to launch a food pantry providing weekly rations to 50+ unhoused individuals in Durham
- Launched the Interfaith Lunch series to support cross-faith understanding, drawing 25+ people weekly

SKILLS

Certifications: AWS Solutions Architect (2024)

Cloud/Infrastructure: AWS, Fargate, DynamoDB, Kafka, CloudWatch

Data/ML: Databricks, Snowflake, TensorFlow, PyTorch

Programming: Python, Java, SQL, Swift, C, C++, MATLAB, Verilog, Assembly, VB.Net

PERSONAL PROJECTS

Agricultural Crop Protection System **Jan 2023 — May 2023**

- Designed a solar-powered crop-protection system that detected animals via a seismic sensor — a novel alternative to camera/motion sensing — & deterred them with automated visual signals generated by a mechanical scarecrow with lights
- Built a calibration model to tune seismic-activity thresholds that distinguished animal's footfall from ambient ground vibration

RISC Processor Project **Aug 2021 — Dec 2021**

- Designed a five-stage 32-bit RISC Processor from scratch to execute MIPS assembly
- Developed an interactive MIPS-based tanks game to run on the custom RISC processor, using flex sensors as input devices